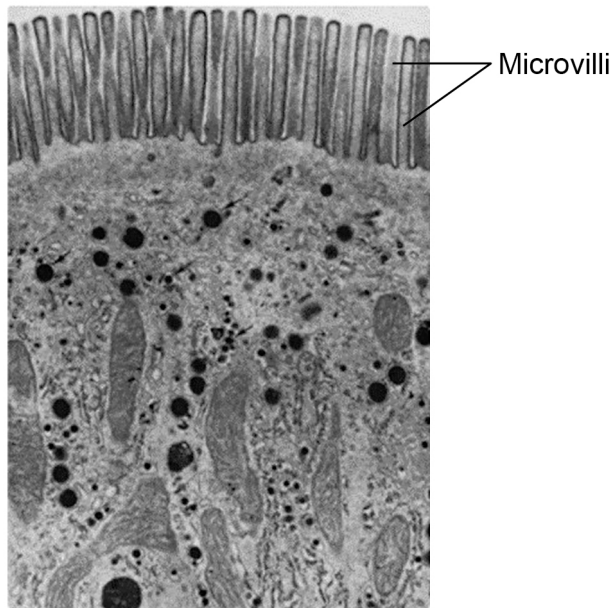


0 3

Figure 1 is an image of part of a cell lining the ileum.

Figure 1



0 3 . 1

Describe and explain **two** ways that structures shown in **Figure 1** increase the rate of absorption by this cell.

[2 marks]

1 _____

2 _____



0 3 . 2

An ileum cell **without** microvilli has $7.85 \mu\text{m}^2$ of cell surface in contact with digested substances.

A scientist found an ileum cell with 1000 microvilli. The microvilli cover the entire cell surface in contact with digested substances.

Microvilli are $0.1 \mu\text{m}$ in diameter and $1 \mu\text{m}$ in length.

The surface area of a microvillus is calculated using this equation

$$2 \pi r l + \pi r^2$$

where π is 3.14,

r is the radius,

l is the length.

Calculate the ratio of the area of the ileum cell surface **with** microvilli to the area of ileum cell surface **without** microvilli.

Show your working.

[2 marks]

Ratio _____:1

Question 3 continues on the next page

Turn over ►



0 3 . 3 The protein ZO-1 is found on the surface of ileum cells.

A scientist used an anti-ZO-1 monoclonal antibody to identify ileum cells in a sample of intestine observed using an optical microscope.

Suggest how the monoclonal antibody helped the scientist to identify ileum cells in the sample of intestine.

[3 marks]

7

